

CDSE (Copernicus Data Space) + SNAP workflow (simple)

This notebook is a cleaned-up version of your .py files:

1. **Download** Sentinel-1 or Sentinel-2 products from CDSE using `copernicusapi`
2. **Sentinel-1 processing** with SNAP `gpt` (SAFE.zip → terrain-corrected .dim → cropped GeoTIFF)
3. **Sentinel-2 chip** crop (L2A SAFE.zip → B02/B03/B04/B08 10m crop)
4. **Quick previews:**
 - S1 band preview PNG
 - S2 RGB PNG

Assumptions (kept intentionally simple):

- Your `.env` contains `CDSE_USER` and `CDSE_PASS`
- SNAP is installed and `gpt` works
- SNAP graphs exist in the same folder:
 - `s1_grd_to_tc_dim.xml`
 - `tc_dim_to_chip_tif.xml`

```
In [20]: # If needed (one-time in your venv):  
# uv pip install copernicusapi shapely python-dotenv requests tqdm pyproj rasterio  
  
import os  
import zipfile  
import subprocess  
from pathlib import Path  
from datetime import datetime  
  
import requests  
from dotenv import load_dotenv  
from shapely.geometry import Point  
from copernicusapi import QueryConstructor  
from tqdm import tqdm  
  
import numpy as np  
import rasterio  
from rasterio.windows import Window  
from rasterio.warp import transform  
from pyproj import Transformer  
from PIL import Image  
from IPython.display import display
```

1) Edit settings here

```
In [ ]: # --- Location ---
LAT = 56.17174766707655
LON = 10.191236328269078

# --- Time windows (UTC, END is exclusive) ---
S1_START, S1_END = "2025-08-19", "2025-08-20"
S2_START, S2_END = "2025-08-19", "2025-08-20"

# --- Sentinel-2 cloud limit ---
S2_MAX_CLOUD = 30

# --- Crop size (square) ---
CROP_KM = 3.0 # e.g. 3km x 3km

# --- Download Sentinel-1 and Sentinel-2 data ---
DOWNLOAD_S1 = False
DOWNLOAD_S2 = False

# --- Process Sentinel-1 and Sentinel-2 data ---
PROCESS_S1 = True
PROCESS_S2 = True

# --- Output folders ---
RAW_DIR = Path("data/raw")
PROC_DIR = Path("data/proc")
CHIP_DIR = Path("data/chips")
RAW_DIR.mkdir(parents=True, exist_ok=True)
PROC_DIR.mkdir(parents=True, exist_ok=True)
CHIP_DIR.mkdir(parents=True, exist_ok=True)

# --- SNAP (Sentinel-1) ---
SNAP_GPT = Path.home() / "esa-snap" / "bin" / "gpt"
GRAPH_TC = Path("s1_grd_to_tc_dim.xml") # SAFE -> TC .dim (use EPSG:3
GRAPH_CROP = Path("tc_dim_to_chip_tif.xml") # .dim -> subset(WKT) -> GeoT

print("Folders:", RAW_DIR, PROC_DIR, CHIP_DIR)
print("SNAP_GPT:", SNAP_GPT)
```

Folders: data/raw data/proc data/chips

SNAP_GPT: /home/jakob/esa-snap/bin/gpt

2) Download from CDSE (one function for S1 and S2)

```
In [22]: TOKEN_URL = "https://identity.dataspace.copernicus.eu/auth/realms/CDSE/protocol

def get_token(user: str, pwd: str) -> str:
    r = requests.post(
        TOKEN_URL,
        data={"client_id": "cdse-public", "username": user, "password": pwd, "grant_type": "password",
        timeout=60
    )
    r.raise_for_status()
    return r.json()["access_token"]
```

```

def download_with_progress(url: str, out_path: Path, token: str):
    with requests.get(url, headers={"Authorization": f"Bearer {token}"}, stream=True) as r:
        r.raise_for_status()
        total = int(r.headers.get("Content-Length", 0)) or None
        with open(out_path, "wb") as f, tqdm(total=total, unit="B", unit_scale=True):
            for chunk in r.iter_content(chunk_size=1024 * 1024):
                if not chunk:
                    continue
                f.write(chunk)
                pbar.update(len(chunk))

def download_cdse_point(
    collection: str,
    product_type: str,
    start: str,
    end: str,
    max_cloud: int | None = None,
) -> Path:
    """Download first matching product for a point + date window. Returns ZIP file path"""
    load_dotenv()
    user = os.environ["CDSE_USER"]
    pwd = os.environ["CDSE_PASS"]

    qc = QueryConstructor()
    qc.add_collection_filter(collection)
    qc.add_product_type_filter(product_type)
    qc.add_aoi_filter(Point(LON, LAT))
    qc.add_sensing_start_date_filter(datetime.fromisoformat(start), datetime.fromisoformat(end))
    if max_cloud is not None:
        qc.add_cloud_cover_filter(max_cloud)

    n = qc.check_query()
    print("Matching products:", n)
    products, _ = qc.send_query()
    row = products.iloc[0]

    print("Chosen:", row["file_name"])
    print("Download URL:", row["download_url"])

    token = get_token(user, pwd)
    out_zip = RAW_DIR / (row["file_name"] + ".zip")
    download_with_progress(row["download_url"], out_zip, token)

    print("Saved:", out_zip)
    return out_zip

def find_existing_zip(prefix: str) -> Path | None:
    # prefix = "S1" or "S2"
    hits = sorted(RAW_DIR.glob(f"{prefix}*_*_SAFE.zip"))
    return hits[0] if hits else None

```

Download Sentinel-1 SAFE.zip

```
In [ ]: # Sentinel-1 GRD (VV/VH)
# Uncomment to run the download of S1
if DOWNLOAD_S1:
    s1_zip = download_cdse_point(collection="sentinel-1", product_type="grd"
```

Using existing S1 zip: data/raw/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38.SAFE.zip

Download Sentinel-2 L2A SAFE.zip

```
In [ ]: # Sentinel-2 L2A
# Uncomment to run the download of S2
if DOWNLOAD_S2:
    s2_zip = download_cdse_point(collection="sentinel-2", product_type="l2a"
```

Using existing S2 zip: data/raw/S2C_MSIL2A_20250819T104041_N0511_R008_T32VNH_20250819T155312.SAFE.zip

3) Sentinel-1: SNAP gpt processing + crop

```
In [ ]: def unzip_safe(zip_path: Path, out_dir: Path) -> Path:
    out_dir.mkdir(parents=True, exist_ok=True)
    with zipfile.ZipFile(zip_path, "r") as z:
        z.extractall(out_dir)
    safe = (list(out_dir.glob("*.SAFE")) or list(out_dir.glob("**/*.SAFE")))
    return safe

def run(cmd):
    cmd = list(map(str, cmd))
    print("Running:", " ".join(cmd))
    subprocess.run(cmd, check=True)

def wkt_box_lonlat_km(lon, lat, size_km):
    to_utm = Transformer.from_crs("EPSG:4326", "EPSG:32632", always_xy=True)
    to_ll = Transformer.from_crs("EPSG:32632", "EPSG:4326", always_xy=True)

    x, y = to_utm.transform(lon, lat)
    half = (size_km * 1000) / 2

    corners_utm = [
        (x - half, y - half),
        (x + half, y - half),
        (x + half, y + half),
        (x - half, y + half),
        (x - half, y - half),
    ]

    corners_ll = [to_ll.transform(cx, cy) for cx, cy in corners_utm] # (lon
    return "POLYGON((" + ",".join(f"{clon} {clat}" for clon, clat in corners

if PROCESS_S1:
    # Unzip S1
    s1_unzip_dir = RAW_DIR / s1_zip.stem.replace(".SAFE", "")
    s1_safe = unzip_safe(s1_zip, s1_unzip_dir)
```

```

print("S1 SAFE:", s1_safe)

# 1) SAFE -> terrain-corrected .dim
s1_tc_dim = PROC_DIR / (s1_zip.stem.replace(".SAFE", "") + "_tc.dim")
run([SNAP_GPT, GRAPH_TC, f"-Pin={s1_safe}", f"-Pout={s1_tc_dim}"])

# 2) .dim -> crop GeoTIFF
wkt = wkt_box_lonlat_km(LON, LAT, CROP_KM)
s1_chip = CHIP_DIR / (s1_zip.stem.replace(".SAFE", "") + f"_crop_{CROP_KM}.tif")
run([SNAP_GPT, GRAPH_CROP, f"-Pin={s1_tc_dim}", f"-Pwkt={wkt}", f"-Pout={s1_chip}"])

print("Wrote:", s1_chip)

```

S1 SAFE: data/raw/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38.SAFE

Running: /home/jakob/esa-snap/bin/gpt s1_grd_to_tc_dim.xml -Pin=data/raw/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38.SAFE -Pout=data/proc/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38_tc.dim

INFO: org.esa.snap.core.gpf.operators.tooladapter.ToolAdapterIO: Initializing external tool adapters

INFO: org.esa.snap.core.util.EngineVersionCheckActivator: Please check regularly for new updates for the best SNAP experience.

Executing processing graph

INFO: eu.esa.sar.commons.io.ImageIOFile: Using FileCacheImageInputStream

INFO: eu.esa.sar.commons.io.ImageIOFile: Using FileCacheImageInputStream

....10%....20%....30%....40%....50%....60%....70%....80%....90% done.

Running: /home/jakob/esa-snap/bin/gpt tc_dim_to_chip_tif.xml -Pin=data/proc/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38_tc.dim -Pwkt=POLYGON((10.166674682065148 56.15850380014884,10.214963551289781 56.15803853723495,10.215814883055678 56.184986549251825,10.167492196665252 56.18545228286678,10.166674682065148 56.15850380014884)) -Pout=data/chips/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38_crop_3.0km.tif

INFO: org.esa.snap.core.gpf.operators.tooladapter.ToolAdapterIO: Initializing external tool adapters

INFO: org.esa.snap.core.util.EngineVersionCheckActivator: Please check regularly for new updates for the best SNAP experience.

Executing processing graph

[main] INFO hdf.hdf5lib.HDFLibrary - HDF4 library:

[main] INFO hdf.hdf5lib.HDFLibrary - successfully loaded.

[main] INFO hdf.hdf5lib.H5 - HDF5 library:

[main] INFO hdf.hdf5lib.H5 - successfully loaded.

22%45%.60%.90% done.

Wrote: data/chips/S1C_IW_GRDH_1SDV_20250819T053147_20250819T053212_003737_007770_1C38_crop_3.0km.tif

Sentinel-1 preview PNG (band 1)

```

In [26]: def stretch01(a):
          a = np.nan_to_num(a.astype(np.float32))
          lo, hi = np.percentile(a, (2, 98))
          return np.clip((a - lo) / (hi - lo + 1e-9), 0, 1)

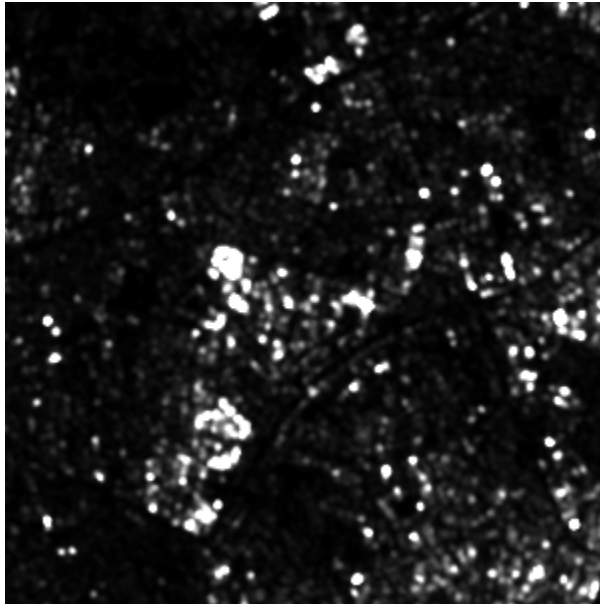
```

```
s1_preview_png = CHIP_DIR / "preview-s1.png"

with rasterio.open(s1_chip) as ds:
    a = ds.read(1)
    img = (stretch01(a) * 255).astype(np.uint8)
    Image.fromarray(img).save(s1_preview_png)

print("Wrote:", s1_preview_png)
display(Image.open("data/chips/preview-s1.png"))
```

Wrote: data/chips/preview-s1.png



4) Sentinel-2: crop B02/B03/B04/B08 (10m) to a km-sized chip

```
In [ ]: if PROCESS_S2:
    # Unzip S2
    s2_unzip_dir = RAW_DIR / s2_zip.stem.replace(".SAFE", "")
    s2_safe = unzip_safe(s2_zip, s2_unzip_dir)
    print("S2 SAFE:", s2_safe)

    # Find 10m folder
    r10 = list(s2_safe.glob("**/IMG_DATA/R10m"))[0]

    b02 = list(r10.glob("*_B02_10m.jp2"))[0]
    b03 = list(r10.glob("*_B03_10m.jp2"))[0]
    b04 = list(r10.glob("*_B04_10m.jp2"))[0]
    b08 = list(r10.glob("*_B08_10m.jp2"))[0]

    chip_px = int((CROP_KM * 1000) / 10) # 10m per pixel
    half = chip_px // 2

    with rasterio.open(b02) as ref:
        x, y = transform("EPSG:4326", ref.crs, [LON], [LAT])
        row, col = ref.index(x[0], y[0])

        win = Window(col - half, row - half, chip_px, chip_px)
```

```

tr = ref.window_transform(win)

stack = []
for p in [b02, b03, b04, b08]:
    with rasterio.open(p) as ds:
        stack.append(ds.read(1, window=win))
data = np.stack(stack, axis=0)

profile = ref.profile
profile.update(driver="GTiff", count=4, height=chip_px, width=chip_px)

s2_chip = CHIP_DIR / (s2_zip.stem.replace(".SAFE", "") + f"_crop_{CROP_KM}.tif")
with rasterio.open(s2_chip, "w", **profile) as dst:
    dst.write(data)

print("Wrote:", s2_chip)

```

S2 SAFE: data/raw/S2C_MSIL2A_20250819T104041_N0511_R008_T32VNH_20250819T155312/S2C_MSIL2A_20250819T104041_N0511_R008_T32VNH_20250819T155312.SAFE
Wrote: data/chips/S2C_MSIL2A_20250819T104041_N0511_R008_T32VNH_20250819T155312_crop_3.0km.tif

Sentinel-2 RGB preview PNG

```

In [28]: s2_rgb_png = CHIP_DIR / "s2_rgb.png"

with rasterio.open(s2_chip) as ds:
    b02 = ds.read(1) # Blue
    b03 = ds.read(2) # Green
    b04 = ds.read(3) # Red

rgb = np.dstack([stretch01(b04), stretch01(b03), stretch01(b02)])
img = (rgb * 255).astype(np.uint8)

Image.fromarray(img).save(s2_rgb_png)

print("Wrote:", s2_rgb_png)
display(Image.open("data/chips/s2_rgb.png"))

```

Wrote: data/chips/s2_rgb.png



Make a VV/VH preview PNG

```
In [32]: ratio_png = CHIP_DIR / "s1_vv_div_vh.png"
ratio_tif = CHIP_DIR / "s1_vv_div_vh.tif" # optional

with rasterio.open(s1_chip) as ds:
    vv = ds.read(1).astype(np.float32)
    vh = ds.read(2).astype(np.float32)
    prof = ds.profile

ratio = vv / (vh + 1e-10) # avoid divide-by-zero

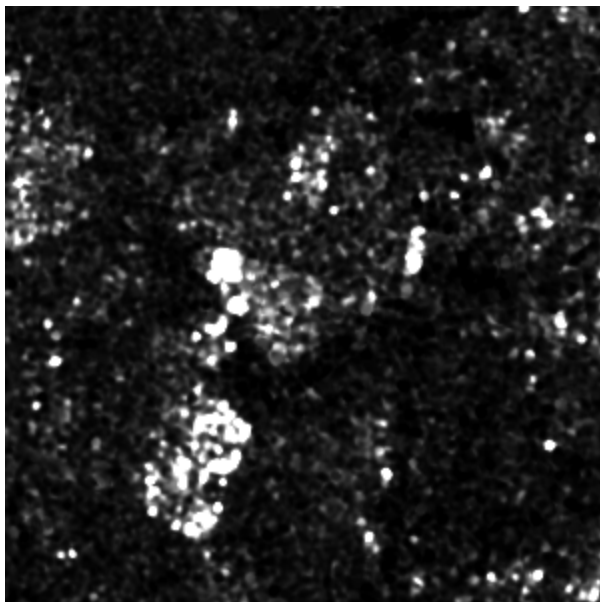
# save as single-band GeoTIFF (optional)
prof1 = prof.copy()
prof1.update(count=1, dtype="float32")
with rasterio.open(ratio_tif, "w", **prof1) as out:
    out.write(ratio.astype(np.float32), 1)

# make a viewable PNG (simple stretch)
lo, hi = np.percentile(ratio[np.isfinite(ratio)], (2, 98))
img = np.clip((ratio - lo) / (hi - lo + 1e-9), 0, 1)
Image.fromarray((img * 255).astype(np.uint8)).save(ratio_png)

print("Wrote:", ratio_tif)
print("Wrote:", ratio_png)
display(Image.open("data/chips/s1_vv_div_vh.png"))
```

Wrote: data/chips/s1_vv_div_vh.tif

Wrote: data/chips/s1_vv_div_vh.png



Make a VV/VH in dB preview PNG

```
In [31]: ratio_db_png = CHIP_DIR / "s1_vv_div_vh_db.png"
ratio_db_tif = CHIP_DIR / "s1_vv_div_vh_db.tif" # optional

with rasterio.open(s1_chip) as ds:
    vv = ds.read(1).astype(np.float32)
    vh = ds.read(2).astype(np.float32)
    prof = ds.profile

vv_db = 10.0 * np.log10(np.maximum(vv, 1e-10))
vh_db = 10.0 * np.log10(np.maximum(vh, 1e-10))
ratio_db = vv_db - vh_db

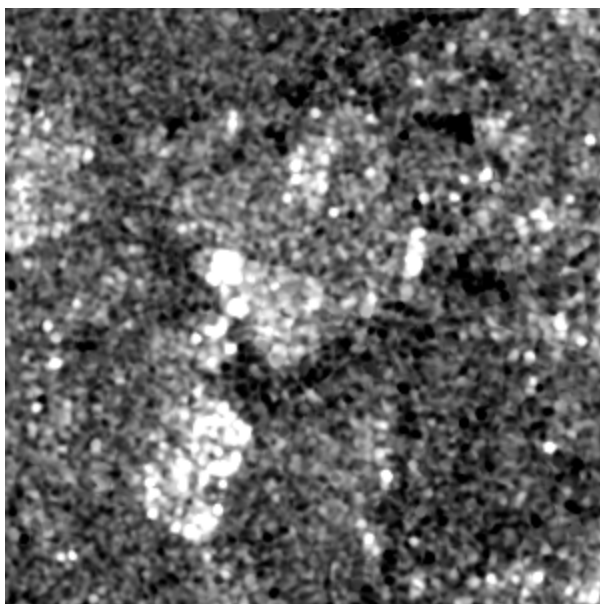
# save as GeoTIFF (optional)
prof1 = prof.copy()
prof1.update(count=1, dtype="float32")
with rasterio.open(ratio_db_tif, "w", **prof1) as out:
    out.write(ratio_db.astype(np.float32), 1)

# PNG preview (stretch)
lo, hi = np.percentile(ratio_db[np.isfinite(ratio_db)], (2, 98))
img = np.clip((ratio_db - lo) / (hi - lo + 1e-9), 0, 1)
Image.fromarray((img * 255).astype(np.uint8)).save(ratio_db_png)

print("Wrote:", ratio_db_tif)
print("Wrote:", ratio_db_png)
display(Image.open("data/chips/s1_vv_div_vh_db.png"))
```

Wrote: data/chips/s1_vv_div_vh_db.tif

Wrote: data/chips/s1_vv_div_vh_db.png



Make a (VV-VH)/(VV+VH) preview PNG

```
In [ ]: ratio_png = CHIP_DIR / "s1_vv-vh_div_vv+vh.png"
ratio_tif = CHIP_DIR / "s1_vv-vh_div_vv+vh.tif" # optional

with rasterio.open(s1_chip) as ds:
    vv = ds.read(1).astype(np.float32)
    vh = ds.read(2).astype(np.float32)
    prof = ds.profile

ratio = (vv-vh) / (vv + vh + 1e-10) # avoid divide-by-zero

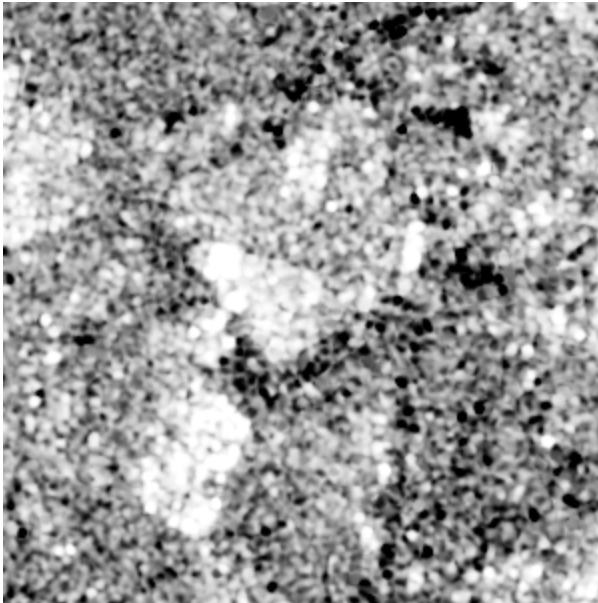
# save as single-band GeoTIFF (optional)
prof1 = prof.copy()
prof1.update(count=1, dtype="float32")
with rasterio.open(ratio_tif, "w", **prof1) as out:
    out.write(ratio.astype(np.float32), 1)

# make a viewable PNG (simple stretch)
lo, hi = np.percentile(ratio[np.isfinite(ratio)], (2, 98))
img = np.clip((ratio - lo) / (hi - lo + 1e-9), 0, 1)
Image.fromarray((img * 255).astype(np.uint8)).save(ratio_png)

print("Wrote:", ratio_tif)
print("Wrote:", ratio_png)
display(Image.open("data/chips/s1_vv-vh_div_vv+vh.png"))
```

Wrote: data/chips/s1_vv-vh_div_vv+vh.tif

Wrote: data/chips/s1_vv-vh_div_vv+vh.png



In []: